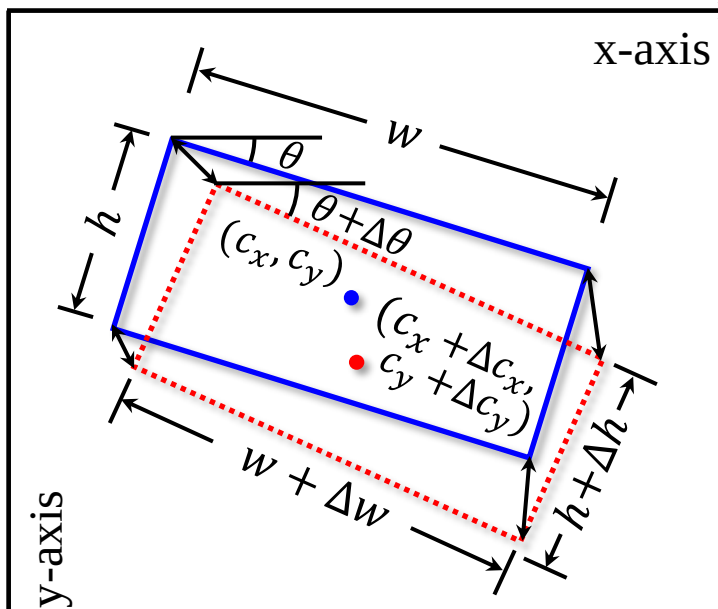
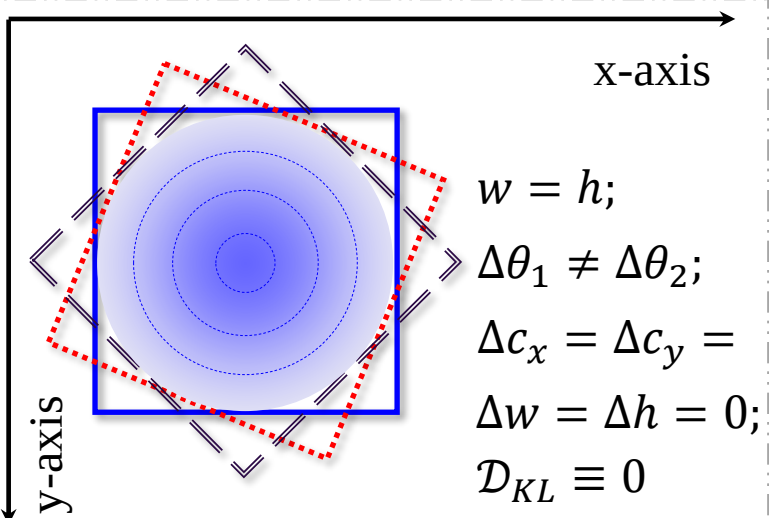
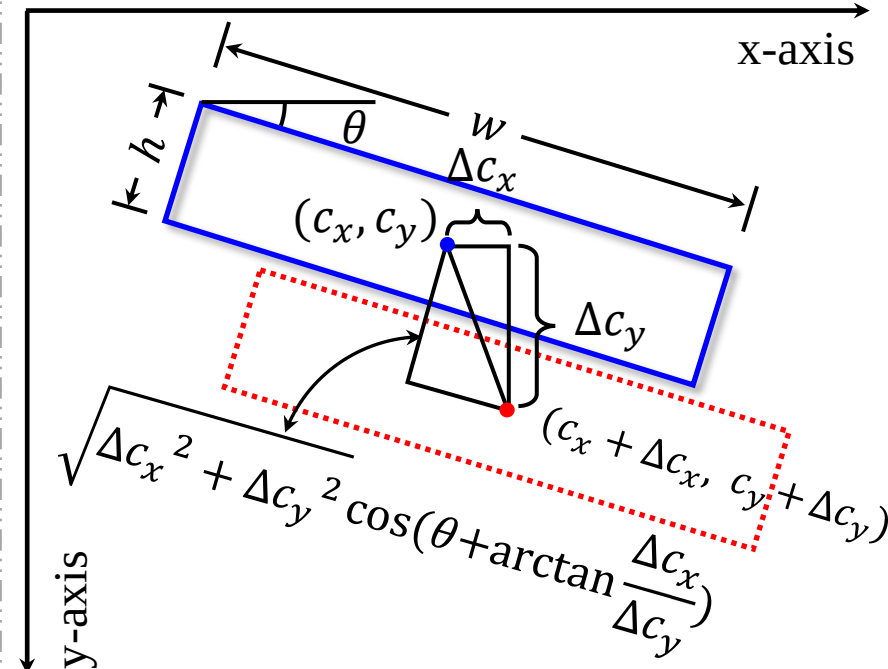




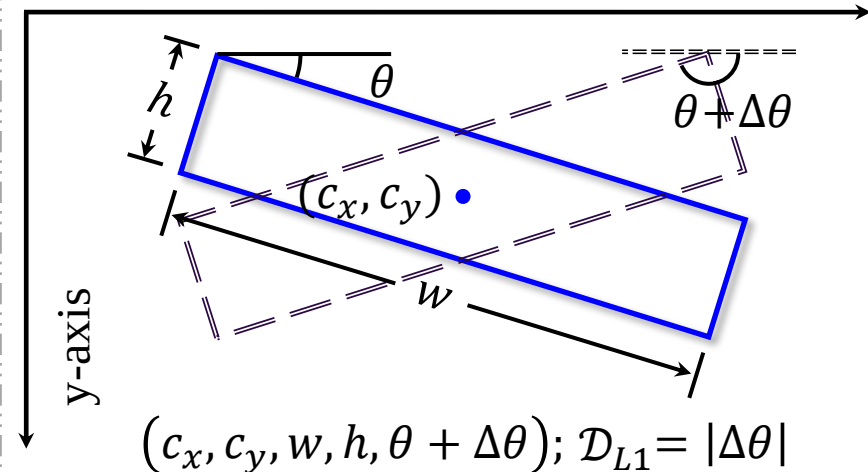
(a) Chamfer distance (ours)



(b) Kullback-Leibler Divergence

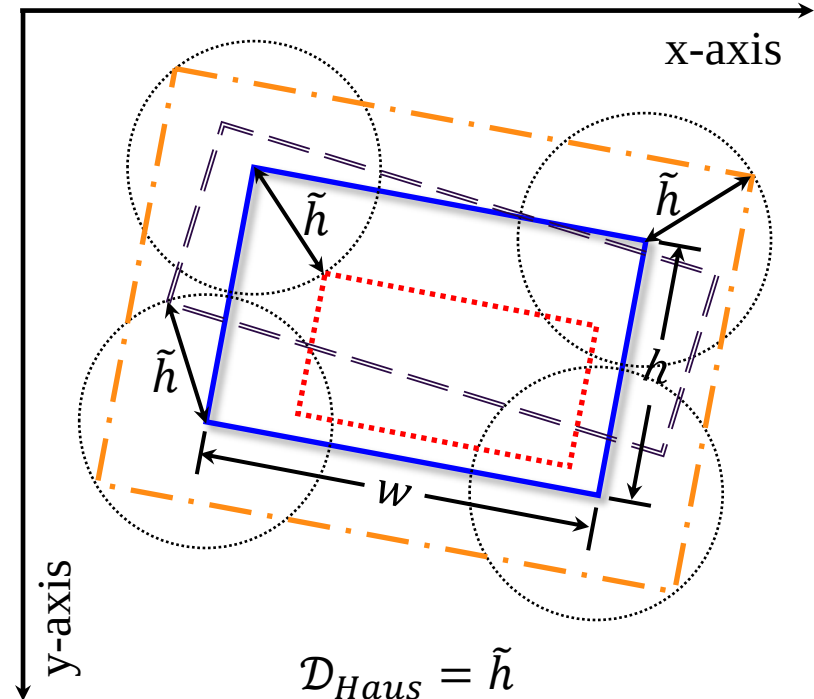


$$(c_x + \Delta c_x, c_y + \Delta c_y, w, h, \theta); \mathcal{D}_{L1} = |\Delta c_x| + |\Delta c_y|$$



$$(c_x, c_y, w, h, \theta + \Delta \theta); \mathcal{D}_{L1} = |\Delta \theta|$$

(c) L1 distance



$$\mathcal{D}_{Haus} = \tilde{h}$$

(d) Hausdorff distance

Ground truth oriented boxes (c_x, c_y, w, h, θ)

Predicted oriented boxes $(c_x + \Delta c_x, c_y + \Delta c_y, w + \Delta w, h + \Delta h, \theta + \Delta \theta)$